

490-March6

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2023-03-06

1

```
library(meta)
```

```
## Warning: package 'meta' was built under R version 4.2.2
```

```
## Loading 'meta' package (version 6.1-0).  
## Type 'help(meta)' for a brief overview.  
## Readers of 'Meta-Analysis with R (Use R!)' should install  
## older version of 'meta' package: https://tinyurl.com/dt4y5drs
```

```
n1 = c(13,24,20,8,21,18)  
n2 = c(150,24,20,28,21,18)  
mean1 = c(150,213,191,356,109,73.1)  
mean2 = c(299,249,185,267,97,59.2)  
sd1 = c(55,136,112,151,47,43.5)  
sd2 = c(55,115,160,267,97,59.2)  
# a.  
meta = metacont(n1, mean1, sd1, n2, mean2, sd2, sm="MD", method.tau="DL")  
cat('Entire print output: \n')
```

```
## Entire print output:
```

```
print(meta)
```

```
## Number of studies combined: k = 6  
## Number of observations: o = 365  
##  
##              MD              95%-CI      z  p-value  
## Common effect model -50.0586 [-69.1280; -30.9891] -5.15 < 0.0001  
## Random effects model -18.1875 [-94.7959;  58.4210] -0.47  0.6417  
##  
## Quantifying heterogeneity:  
## tau^2 = 7816.7695 [1757.8429; 46120.7395]; tau = 88.4125 [41.9266; 214.7574]  
## I^2 = 92.3% [85.9%; 95.8%]; H = 3.60 [2.67; 4.85]  
##  
## Test of heterogeneity:
```

```
##      Q d.f.  p-value
## 64.70    5 < 0.0001
##
## Details on meta-analytical method:
## - Inverse variance method
## - DerSimonian-Laird estimator for tau^2
## - Jackson method for confidence interval of tau^2 and tau
```

```
cat('Summary output: \n')
```

```
## Summary output:
```

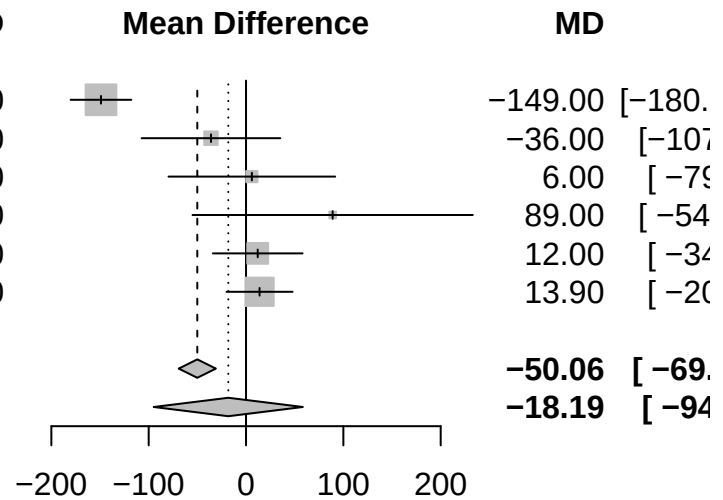
```
summary(meta)
```

```
##      MD      95%-CI %W(common) %W(random)
## 1 -149.0000 [-180.1664; -117.8336]      37.4      18.9
## 2  -36.0000 [-107.2551;  35.2551]       7.2      16.7
## 3   6.0000 [ -79.5946;  91.5946]       5.0      15.7
## 4   89.0000 [ -54.9761; 232.9761]       1.8      11.6
## 5   12.0000 [ -34.1003;  58.1003]      17.1      18.3
## 6   13.9000 [ -20.0378;  47.8378]      31.6      18.8
##
## Number of studies combined: k = 6
## Number of observations: o = 365
##
##      MD      95%-CI      z  p-value
## Common effect model -50.0586 [-69.1280; -30.9891] -5.15 < 0.0001
## Random effects model -18.1875 [-94.7959;  58.4210] -0.47  0.6417
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## Test of heterogeneity:
##      Q d.f.  p-value
## 64.70    5 < 0.0001
##
## Details on meta-analytical method:
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## - Jackson method for confidence interval of tau^2 and tau
```

```
# b.
forest(meta)
```

Experimental Mean	SD	Total	Mean	Control SD
150.00	55.0000	150	299.00	55.0000
24.00	136.0000	24	249.00	115.0000
20.00	112.0000	20	185.00	160.0000
28.00	151.0000	28	267.00	267.0000
21.00	47.0000	21	97.00	97.0000
18.10	43.5000	18	59.20	59.2000
		261		

95, $p < 0.01$



#2

```
# a. effect size estimate
theta = mean1 - mean2
# b. variance of effect size
vtheta = sd1^2/n1 + sd2^2/n2
# c. 95% CI for ES
errtheta = sqrt(vtheta)
alpha = 0.05
UCL = theta + qnorm(1-alpha/2)*(errtheta)
LCL = theta - qnorm(1-alpha/2)*(errtheta)
cat('95% confidence intervals for mean difference estimates: \n', LCL, UCL, '\n')
```

95% confidence intervals for mean difference estimates:

-180.1664 -107.2551 -79.59457 -54.97614 -34.10034 -20.03779 -117.8336 35.25509 91.59457 232.9761 58

```
# d. combine all
effsize_data = cbind(theta,vtheta,LCL,UCL)
cat('Combined information: \n')
```

Combined information:

effsize_data

##	theta	vtheta	LCL	UCL
----	-------	--------	-----	-----

```
## [1,] -149.0 252.8590 -180.16645 -117.83355
## [2,] -36.0 1321.7083 -107.25509 35.25509
## [3,] 6.0 1907.2000 -79.59457 91.59457
## [4,] 89.0 5396.1607 -54.97614 232.97614
## [5,] 12.0 553.2381 -34.10034 58.10034
## [6,] 13.9 299.8272 -20.03779 47.83779
```

3 Fixed Effect Model

```
# a.
weights = 1 / vtheta
# b.
weight_sum = sum(weights)
# c.
relative_weights = weights / weight_sum
# d.
M = sum(weights*theta)/weight_sum
# e.
errM = sqrt(1/weight_sum)
M_LCL = M - qnorm(1-alpha/2)*errM
M_UCL = M + qnorm(1-alpha/2)*errM
# f.
M_ts = M / errM
M_pval = 2*(1-pnorm(abs(M_ts)))
cbind(M,M_LCL,M_UCL,M_pval) # extremely significant result
```

```
##           M      M_LCL      M_UCL      M_pval
## [1,] -50.05855 -69.12799 -30.98912 2.674667e-07
```

4 Random Effects Model

```
# a i
Q = sum(weights*theta^2)-(sum(weights*theta))^2/weight_sum
# a ii
C = weight_sum - sum(weights^2)/weight_sum
# a iii
df = 6-1
# a iv
tau_sq = (Q-df)/C
# b
wR = 1 / (vtheta+tau_sq)
# c
total_wR = sum(wR)
# d
rel_wR = wR/sum(wR)
# e
M_R = sum(rel_wR*theta)
# f
```

```

errM_R = sqrt(1/total_wR)
MR_LCL = M_R - qnorm(1-alpha/2)*errM_R
MR_UCL = M_R + qnorm(1-alpha/2)*errM_R
# g
MR_ts = M_R / errM_R
MR_pval = 2*(1-pnorm(abs(MR_ts)))
cbind(rel_wR)

```

```

##          rel_wR
## [1,] 0.1893233
## [2,] 0.1671797
## [3,] 0.1571136
## [4,] 0.1156268
## [5,] 0.1825289
## [6,] 0.1882277

```

```

rbind(M_R,errM_R,MR_LCL,MR_UCL,MR_pval)

```

```

##          [,1]
## M_R      -18.1874599
## errM_R    39.0866769
## MR_LCL   -94.7959389
## MR_UCL    58.4210192
## MR_pval   0.6417088

```

5

```

# The metacont function gives a significant result for the overall mean
# difference using the fixed-effect model, but an insignificant result using
# the random effects model. The results when doing the analysis step-by-step
# are identical. The discrepancy between fixed/random effects is probably due to
# the fact the results of the study with the most weight (i.e. lowest within
# variance), contradict the results of the studies with the 2nd and 3rd highest
# weights, pulling the overall mean difference estimate closer to 0. Many of the
# studies in the meta-analysis have very high variances, and it is clear that
# tau^2 (or between-study variability) is high, leading to a wider confidence
# interval around M in the random effects model. This variability leads to the
# confidence interval stretching over 0, suggesting there is no significant
# difference between the two groups. This is further supported by the large
# p-value obtained in 4g.
# Since the heterogeneity test yield a high p-value, it can be concluded that
# the tau^2 value is significantly different to zero. In other words, there is
# significant variability between studies, essentially nullifying the results
# obtained from the fixed effect model. Fixed-effect gave a significant result
# for mean difference because it operates on the assumption that the only source
# of error is from sampling. Since random-effects has two sources of error
# (variability), it is guaranteed to have at least the same total variability.
# In this case, random-effects shows there is between-study variability, raising
# total variability, and in turn leading to a wider confidence interval and
# larger p-value. This is to say: researchers may conclude that there is no
# significant difference between the means of the two groups.

```